LEAKLISTEN · DESIGN DECISION RECORD

LeakListen TRL 2 review decisions

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DOCUMENT CONTROL

Document ID	LKL-DDR-001	Version	0.3
Status	Draft	Date	2026-10-02
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/leaklisten		
Source	docs/decisions/0001-trl2-review-decisions.md @ a81e84a		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	O1 and O2 decided by Amish on 2026-10-02 (LKL-DEC-001, items 4 and 5)
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002): D1 to D6 decided; O1 and O2 still open
0.1	2026-09-25	Amish Chadha	Record the TRL 2 review items adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** accepted for D1 to D6. On 2026-09-25 Amish wrote "i accept all your recommendations, go with them across all repos", so D1 to D6 are "Decided by Amish, 2026-09-25: go with recommendation" (see LKL-DDR-002). Items O1 and O2 had no recommendation and remain "Proposed, awaiting Amish". O1 and O2 were decided by Amish on 2026-10-02 as recommended in the design decisions register (LKL-DEC-001, items 4 and 5): "i approve your recommendations for all 555 open decisions."

Context

The TRL 2 review note (`docs/REVIEW.md`, session 2026-09-25, /populate) listed eight items as "Proposed, awaiting Amish". On 2026-09-25 Amish asked for this batch of repos to be taken through the usual process with the instruction "you know the drill, nothing gets past TRL 3". He has not reviewed this repo's items one by one. Every item that carried a recommendation is therefore adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review. Items without a recommendation stay open. Version 0.1 of this record contained no decision made by Amish; version 0.2 records his acceptance of D1 to D6 on 2026-09-25.

Options considered

The options for each item are those listed in `docs/REVIEW.md` (session 2026-09-25, /populate) and in LKL-PRC-001 v0.2, Key design choices.

Decision

Table 1. Items decided.

#	Item	Recommendation (now the decision)	Status
D1	Screening versus correlation (pitch-level)	Option A: nightly noise-level screening only. Time-synchronized correlation between loggers (option B) is recorded as a later variant.	Decided by Amish, 2026-09-25: go with recommendation
D2	Radio from under covers	Option A: flat antenna under the cover, with the TRL 3 link budget deciding whether a through-cover antenna (B) or a nearby gateway (C) is needed. LKL-CAL-001 section D finds that it is needed under iron covers; the response is decided in LKL-DDR-002.	Decided by Amish, 2026-09-25: go with recommendation
D3	Sensor type	Piezo disc with a charge amplifier, with a hydrophone on hydrants kept as an option for plastic networks.	Decided by Amish, 2026-09-25: go with recommendation
D4	Reuse of the FieldNode radio core	Yes: the STM32WL module and payload conventions are shared with FieldNode; FieldNode's solar power is not used.	Decided by Amish, 2026-09-25: go with recommendation
D5	Primary cell size	C size, about 7.7 Ah.	Decided by Amish, 2026-09-25: go with recommendation
D6	Privacy rule	Raw samples never leave the device; R8 is kept as a fixed requirement.	Decided by Amish, 2026-09-25: go with recommendation

Table 2. Items that remain open.

#	Item	Status
O1	First partner utility and region for co-design and a pilot district. No recommendation was made.	Decided by Amish, 2026-10-02 (LKL-DEC-001, item 4): a water utility with metallic mains, district metered areas and a leakage target; first candidate region to approach, the UK (radio band EU868). Nothing is agreed.
O2	Whether a vibration calibration check belongs in LeakListen or in CalRig (CalRig covers temperature, humidity and particles only). No recommendation was made.	Decided by Amish, 2026-10-02 (LKL-DEC-001, item 5): a small shaker bay in CalRig, with LeakListen defining the acceptance test (frequency range, level and pass band) in its own repo.

Consequences

- `project.yaml` : unchanged apart from the TRL fields. No new budget, pitch or problem wording was recommended, so `budget_usd` stays at $130 and the pitch and problem lines are unchanged.
- LKL-PRB-001, LKL-PRC-001 and LKL-REQ-001 are revised to v0.3. The key design choices in the precis are no longer "proposed". R8 is marked as a fixed requirement (D6). No requirement target is relaxed or redefined by these decisions.
- D4 aligns the radio with FieldNode (FND-CAL-001): +14 dBm in EU868 and the same link budget method. LKL-CAL-001 found that the TRL 2 assumption of +20 dBm is not permitted in EU868; the design now uses +14 dBm there and up to +20 dBm in US915.

- The TRL 3 calculations (LKL-CAL-001) led to four design changes within these decisions: a 53 g seismic mass in a 42 mm tall puck, a 24-byte summary, regional transmit power and a weekly network time correction. They add $1 to the parts cost, now $109.00.
- LKL-CAL-001 v0.1 showed R2 and R7 not met and R1, R3, R4 and R9 at risk. The responses (radio option, scope for plastic mains, a larger magnet) were decided by Amish on 2026-09-25 and are recorded in LKL-DDR-002.
- TRL 4 is on hold by Amish's instruction. Nothing in this record authorizes building or testing.